# San Diego Short-Term Residential Occupancy (STRO) Ordinance Analysis

*Group 37 MATH 189 Final Project Report*
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## Introduction

San Diego's rental market is currently navigating an affordability crisis. A 2024 study by researchers at Chapman University designated San Diego as an "impossibly unaffordable" city, ranking it as the fourth most unaffordable in the United States. A widely debated cause of this crisis is the rapid increase of commercial short-term rental units (STRs), which critics argue reduces the local housing supply in popular tourist neighborhoods like La Jolla and Pacific Beach.

This localized concern is grounded in strong international evidence regarding the externalized impact of commercial Airbnbs on housing markets. A study of Barcelona's housing market demonstrated that Airbnb presence raised average neighborhood rents by nearly 2%, with the increase spiking to 7% in neighborhoods with the highest density of listings. Similarly, a study conducted in Berlin found that each additional commercial Airbnb listing displaces 0.23 to 0.37 rental units and increases rent per square meter by 1.3 to 2.4 percent.

In an attempt to combat this crisis locally, the city of San Diego implemented the Short-Term Residential Occupancy (STRO) ordinance in May 2023. This policy capped whole-home short-term rental units at 1% of the city's housing stock, effectively eliminating nearly 50% of the short-term rentals across the city. This policy presented an ideal opportunity for our group to investigate a critical question: did the May 2023 STRO ordinance cause a statistically significant decrease in long-term rent prices, and was this effect stronger in areas with a higher pre-ordinance density of Airbnb listings?

## Hypothesis

Our null hypothesis is that the implementation of short-term rental caps did not have a differential effect on the average Airbnb listing price or long-term rent prices between high-density and low-density areas.

Our alternative hypothesis is that the implementation of short-term rental caps did have a differential effect on the average Airbnb listing price and long-term rent prices between high-density and low-density areas.

## Data Sources & Description

To conduct our analysis of both the short-term and long-term housing markets, we analyzed and combined two separate datasets:

- *InsideAirbnb Data:* We utilized publicly available, point-in-time snapshot data of Airbnb listings in San Diego, directly downloading the CSV files from insideairbnb.com. We selected one pre-ordinance dataset (March 2023) and one post-ordinance dataset (June 2025). We analyzed key features from this datasheet including price, neighborhood, latitude, longitude, and room type.

- *Zillow Observed Rent Index (ZORI)*: To measure the externalized effect on the long-term residential housing market, we imported the longitudinal ZORI data for San Diego ZIP codes, filtering for the corresponding January 2023 and January 2025 timeframes.

## Exploratory Data Analysis (EDA)

Our initial data preprocessing involved isolating “Entire home/apt” properties in the InsideAirbnb dataset, as the STRO ordinance specifically targeted whole-home rentals rather than shared rooms. To avoid post-treatment bias, we classified neighborhoods into “High-Density” and “Low-Density” groups based on their pre-ordinance (March 2023) listing counts. We then mapped these neighborhood densities to the broader ZIP codes used in the ZORI dataset.

From our visualizations in the EDA, overlapping histograms and boxplots revealed severe right-skewness in the Airbnb price distributions, driven by incredibly expensive, luxurious properties. To handle this, we filtered out the top 1% of extreme price outliers (listings priced above $1999 per night). Even after filtering, our Quantile-Quantile (QQ) plots revealed heavy tails. To satisfy the normality assumptions required for linear modeling, we applied a natural logarithm transformation to both the Airbnb listing prices and the ZORI rent prices. We also plotted Empirical Cumulative Distribution Functions (ECDFs) to establish baseline differences in cumulative probability between our density groups prior to regression.

## Statistical Analyses

We applied Multiple Linear Regression using Ordinary Least Squares (OLS) in a Difference-in-Differences framework to evaluate the differential impact of the ordinance.

1. *Short-Term Market Mode*l: We modeled the natural log of Airbnb prices against Density_Group and Time_Period. As our hypothesis test relied on evaluating the interaction term of Density_Group × Time_Period.
2. *Long-Term Market Model*: We ran a secondary OLS model to test the exact same interaction effect on the natural log of ZORI rent prices.
3. *Diagnostics*: For our regression models, we verified assumptions using Cook’s Distance to check for influential outliers and Autocorrelation Function (ACF) plots to ensure residual independence.
4. *Nonparametric Testing*: Since the residuals in our initial Airbnb price model exhibited slight non-normality despite the log transformation, we performed a non-parametric Two-Sample Kolmogorov-Smirnov (KS) test to evaluate if the empirical distribution functions of prices in High-Density and Low-Density areas were statistically distinct.

## Conclusion & Results

In our initial regression model, the interaction term between density and time yielded a coefficient of 0.0297 with a p-value of 0.444. Since this p-value is significantly greater than our alpha-value of $\alpha = 0.05$, we fail to reject the null hypothesis. While the nonparametric KS test confirmed a structural pricing difference exists between high and low-density markets (p-value = 1.82e-15), the regression confirms this gap did not differentially shift due to the policy. Capping the supply did not cause the remaining Airbnbs in highly saturated markets to disproportionately raise their rates compared to less saturated areas.

The ZORI regression yielded an interaction coefficient of -0.0122 with a p-value of 0.930. We fail to reject the null hypothesis. There is no statistical evidence that long-term rents decreased disproportionately in high-density Airbnb ZIP codes after the ordinance was enacted. The influx of former short-term rentals back into the residential market was likely absorbed instantly by San Diego's massive overarching housing deficit.

Our final analysis stayed relatively the same as our initial proposal's methodology of utilizing OLS, log-transformations, QQ-plots, and the KS test. However, we made an addition to our analysis by using the ZORI data. Our proposal's stated goal was to see if the ordinance caused a "decrease in long-term rent prices," but our original data source was exclusively Airbnb listing data. Realizing that Airbnb prices only measure the short-term market, we expanded the scope of our project to incorporate the ZORI dataset, allowing us to answer our problem statement with the necessary statistical analysis.

## Limitations

The primary limitation of our analysis lies in geographic scale. Our ZORI long-term rent dataset was aggregated at the ZIP code level, which is much broader than the highly specific neighborhood and coordinate-level data provided by InsideAirbnb. This discrepancy in spatial resolution may have diluted highly localized rent changes on specific streets or blocks. Additionally, observing market changes over a two-year period introduces confounding macroeconomic variables, such as inflation, changing interest rates, and post-pandemic travel trends, that our model could not fully isolate.

## References


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## GitHub Repository

https://github.com/amitine26/Group37_MATH189_Final/tree/main